

Python Roadmap: Beginner to Advanced

Beginner Level (Basic Python Skills)

- Introduction to Python: What is Python, Installation, First Program
- Python Syntax & Indentation
- Data Types: int, float, str, bool
- Variables and Constants
- Operators: Arithmetic, Comparison, Logical
- Input and Output: input(), print(), f-strings

Intermediate Level (Problem Solving Skills)

- Control Statements: if, elif, else
- Loops: for, while, break, continue, pass
- Collections: Lists, Tuples, Sets, Dictionaries
- Functions: def, return, *args, **kwargs, lambda
- List Comprehension
- Modules and Packages: import, pip
- File Handling: open, read, write, CSV, JSON
- Error Handling: try, except, finally, raise

Python Roadmap: Beginner to Advanced

Advanced Level (Professional Python)

- OOP: Classes, Objects, `__init__`, Inheritance, Polymorphism, Encapsulation
- Iterators and Generators: `__iter__`, `__next__`, `yield`
- Decorators: `@decorator`
- Regex: `re` module for pattern matching
- Lambda, Map, Filter, Reduce
- Datetime Module

Expert/Real-World Python (Projects & Job-Ready Skills)

- APIs and Web Requests: `requests` module
- Web Scraping: `BeautifulSoup`, `Selenium`
- Database: `sqlite3`, `MySQL`
- Virtual Environment: `venv`, `requirements.txt`
- Testing: `unittest`, `pytest`
- Multithreading and Multiprocessing
- Web Development: `Flask`, `Django`, REST APIs
- Data Science & AI: `NumPy`, `Pandas`, `Matplotlib`, `Scikit-learn`, `TensorFlow` (optional)